JJ-B-178b
APRIL 19, 1957
SUPERSEDING
Int. Fed. Spec.
JJ-B-00178a (GSA-FSS)
June 15, 1956, and
Fed. Spec. JJ-B-178
July 6, 1946

FEDERAL SPECIFICATION

BELT, ROUND (DENTAL ENGINES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—This specification covers round, all cord dental-engine belts of various lengths for transmitting power from dental engines to dental handpieces.

1.2 **Classification.**

1.2.1 *Type and lengths.* — Dental-engine belts in accordance with this specification, shall be of but one type and of the lengths specified in the invitation for bids.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 **Specifications and standards.**—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-585—Boxes; Wood, Wirebound.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-676 — Boxes, Set-Up, Paperboard.

Federal Standard:

No. 102—Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-4229 — Boxes; Paperboard, Metal-Stayed.

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

402174/57/No. 825

FSC Class 3030

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Military Standard:

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Thread.*—The thread shall be linen or cotton.

3.2 **Description.**—The belt shall be composed of a sufficient number of threads so interwoven as to form an endless unspliced round belt approximately 1/10 inch in diameter.

3.3 **Length.**—The length of each belt shall be as specified in the invitation for bids, within the plus tolerance shown in table I, when measured under tension as described in 4.3.1.

TABLE I.—*Tolerances*

Belt length	Plus tolerance
Feet	*Inches*
Up to 5	1
5 to 8	1.5
8 to 11	1.75
11 to 15	2
15 to 20	2.25

3.4 **Breaking strength.** — The breaking strength shall be not less than 60 pounds, when tested as specified in 4.3.2.

3.5 **Elongation.**—The stretch shall not exceed 3 percent, when tested as specified in 4.3.3.

3.6 **Workmanship.**—Workmanship shall be first class throughout. The finished belts shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—At least five belts of each length shall be taken at random from each delivery, for inspection and tests.

4.2 **Inspection.**—Samples shall be carefully inspected for material, workmanship, dimensions, and construction.

4.3 Tests.

4.3.1 *Measurement of length.*—Length shall be measured with the belt under a 3-pound load, 1 minute after the application of the load.

4.3.2 *Breaking strength.* — The breaking strength test shall be made on an inclination balance-type machine having smooth grips of the drum type, not less than 1 inch in diameter. The pulling drum shall travel at a uniform rate of 12 inches per minute ± 2 inches, under no load. The distance between the centers of the drum at the beginning of test shall be 10 ± 2 inches. If the specimen breaks less than one-half inch from the point at which it touches the drum or at any point touching the drum, the result shall be disregarded. The test shall be made at an ambent temperature of 70°F. and a relative humidity of 65 plus or minus 2 percent. At least five belts shall be tested for breaking strength with the joint section centered between the grips. The results shall be averaged.

4.3.3 *Elongation.*—The stretch shall be determined by subjecting the entire uncut belt to a 10-pound tension for 24 hours, the measurements being taken before tension is applied and 24 hours after the tension is released. The measurements taken before and after applying the 10-pound load shall be made with the belt under the 3-pound load, as specified in 4.3.1.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue.*—One belt, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table II shall be packaged in each unit, intermediate and exterior container as applicable for the required level of protection specified in procurement document.

TABLE II.—*Packaging quantities*

Federal Stock Number	Unit package	Intermediate package	Shipping container
3030–299–8585	1	12	576
3030–525–4150	1	12	288
3030–542–1550	1	12	288
3030–542–1555	1	12	288

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specifications PPP–B–566, PPP–B–676, or Military Specification MIL–B–4229. Carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size, or a wrapping of kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585; style 8, for class 8 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Notes.—Strapping will not be required for shipping containers forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using class 1 plywood; or an exterior container designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by

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common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Interior package.*—Each unit package and each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color marking and symbol are not required on containers for level A packs when addressed to domestic establishments for the Medical Services of the Armed Forces.

6. NOTES

6.1 Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 **Ordering data.** — Purchasers should specify length required and should exercise any desired options offered herein. (See 1.2.1, 5.1 and 5.2.)

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid samples should be definitely stated, the specification to apply in all other respects.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally procured to meet Federal Government requirements.

6.5 **Transportation description.**—Transportation description applicable to these items is:

Belts, Not Otherwise Indexed by Name.
Carload minimum weight 30,000 pounds.
Motor Volume minimum weight 30,000 pounds.

6.6 This specification covers the following items appearing in the Armed Services Medical Stock List:

3030–299–8585—Belt, Round Engine, 11 Foot.

3030–525–4150—Belt, Round, Foot Pedal Type Engine, 17 Foot, 3 Inch.

3030–542–1550—Belt, Round, Engine, 9 Foot, 4 Inch.

3030–542–1555—Belt, Round, Engine, 9 Foot, 10 Inch.

Patent notice.—When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 10 cents each.

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